

- 10 (a)** The respiratory quotient (RQ) is used to indicate what type of substrate is being metabolised in respiration.

(i) Define the term respiratory quotient.

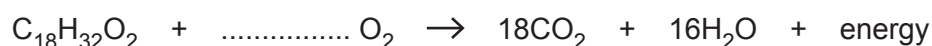
.....

.....

.....

..... [1]

(ii) When the unsaturated fatty acid linoleic acid is respired aerobically the equation is:



Calculate how many molecules of oxygen are used when one molecule of linoleic acid is respired aerobically.

answer [1]

(iii) Calculate the RQ for linoleic acid.

answer [1]

(b) Hummingbirds feed on nectar from flowers. Nectar is rich in sugars.

Fig. 10.1 shows a hummingbird.



Fig. 10.1

A study of aerobic respiration in captive hummingbirds was carried out. The hummingbirds were allowed to feed freely and then made to fast for 4 hours. During the fasting period their RQ values were calculated every 40 minutes.

Fig. 10.2 shows the results from this study.

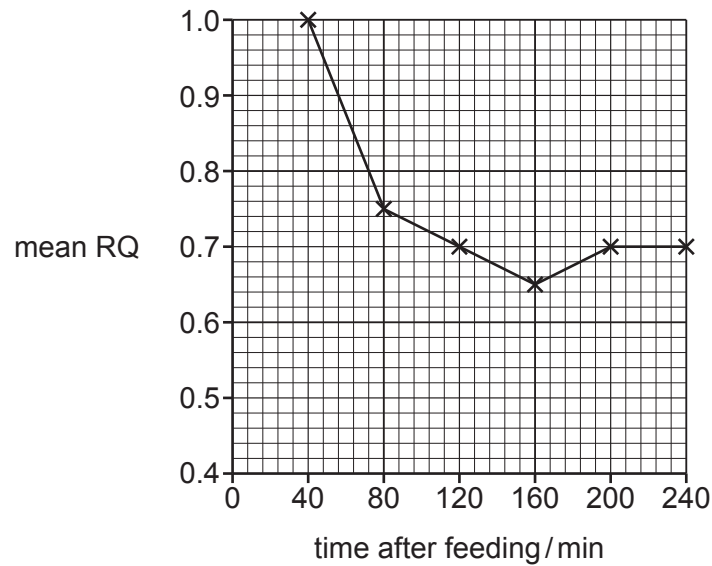


Fig. 10.2

Describe **and** suggest explanations for the results shown in Fig. 10.2.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 7]